

Supplemental information

Class I HDAC inhibition enhances the stem-like memory properties of CRISPR-engineered CAR T cells in neuroblastoma

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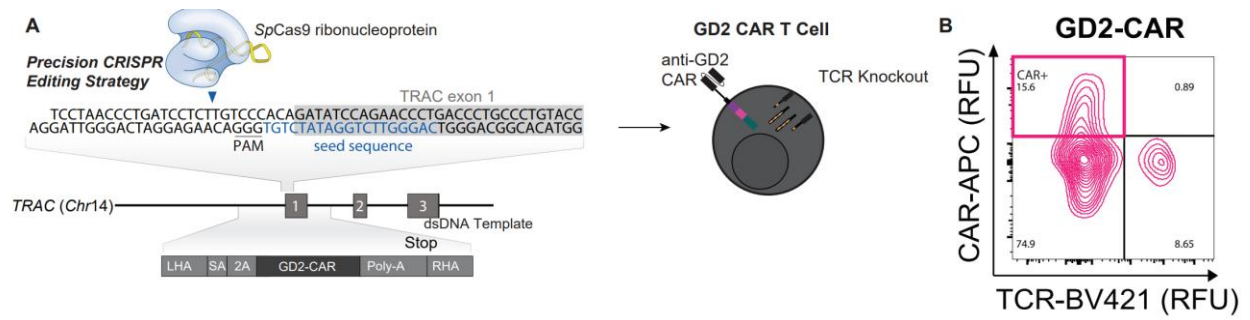


Figure S1. GD2 *TRAC*-CAR T Cell Integration Strategy. (A) A CRISPR/Cas9 ribonucleoprotein was targeted to exon 1 of the *TRAC* locus where a homology-directed repair template containing the GD2 CAR was inserted. The resultant anti-GD2 *TRAC*-CAR-T cell lacked expression of the endogenous TCR. (B) Representative contour plot depicting CAR and low TCR expression as analyzed by flow cytometry.

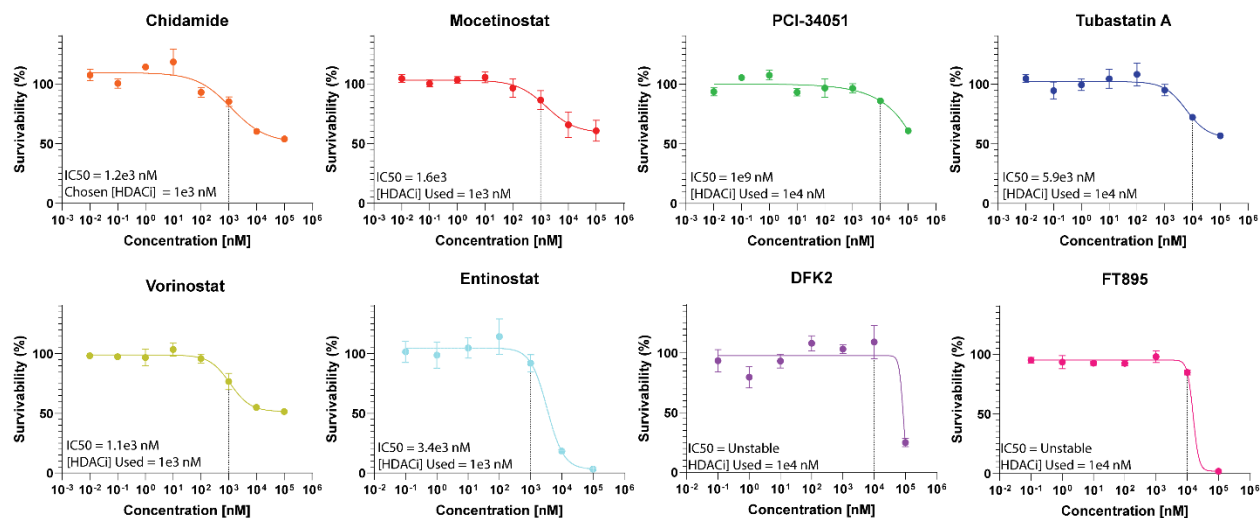


Figure S2. Dosing Curves for HDACi-treated GD2 TRAC-CAR T Cells. GD2 TRAC-CAR T cells were incubated with eight concentrations (10^{-2} - 10^5 nM) of HDAC inhibitors Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat, Entinostat, DFK2, and FT895 for 72 hours upon which the percent of T cells surviving was calculated using a CCK-8 cell counting assay. The curves were fitted to a Sigmoidal 4PL with least squares fit, the IC₅₀ identified, and optimal sub-therapeutic dose found. (One donor, N = 4 data points each) Error bars represent mean and standard error of the mean.

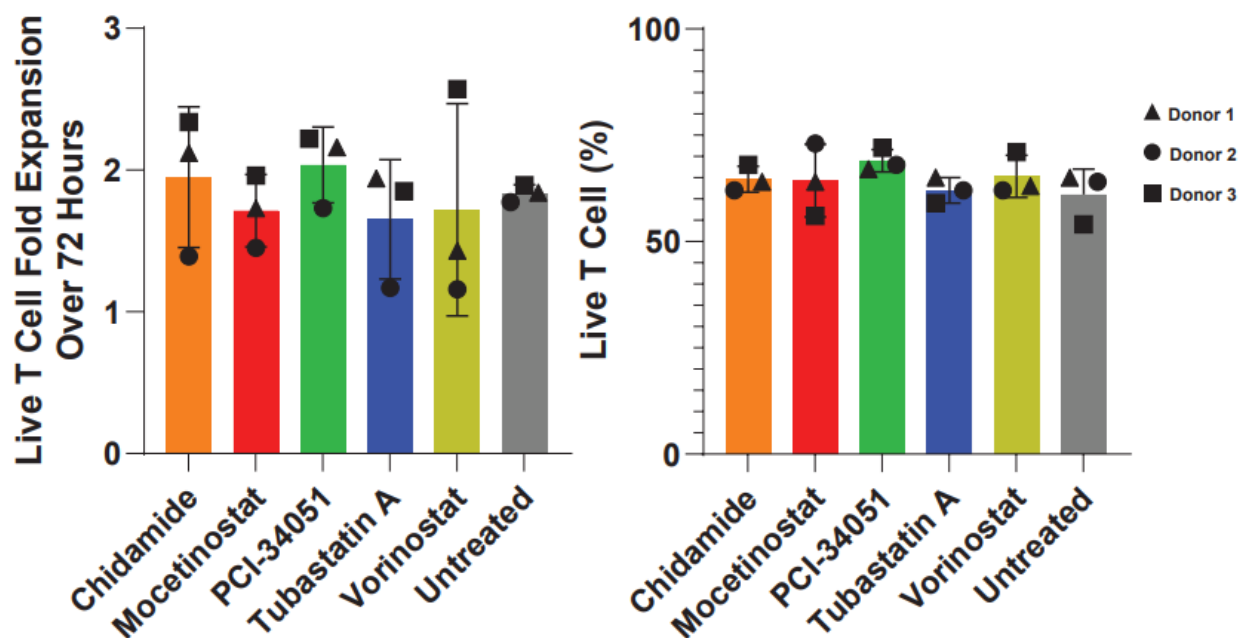


Figure S3. Fold expansion and viability of GD2 *TRAC*-CAR T cells treated with HDAC inhibitors. Bar graphs of the fold expansion and percent of live, thawed GD2 *TRAC*-CAR T cells after a 72-hour treatment with HDAC inhibitors (Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat). Error bars represent standard deviation.

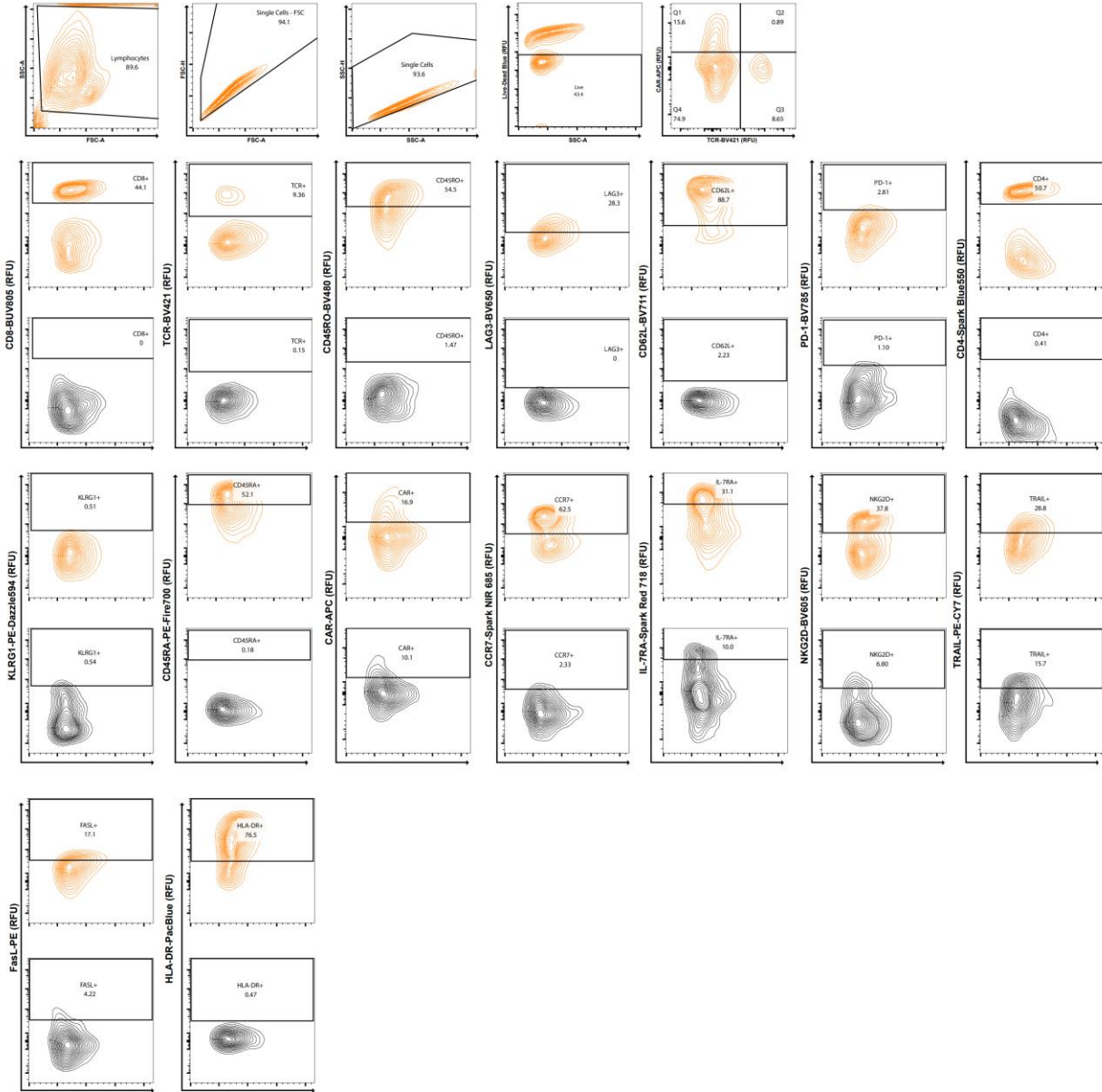


Figure S4. Spectral flow cytometry gating strategy for *in vitro* analysis of GD2 TRAC-CAR T cells. GD2 TRAC-CAR T cells were stained with an optimized antibody panel and analyzed via flow cytometry in FlowJo. Live T cell events were isolated by excluding debris, doublets, and dead cells from the analysis. Positive gates for each surface marker were identified by staining a fluorescence minus one control with each antibody except for the one of interest. The FMO was then used to isolate a positive population in a fully stained sample, both of which are displayed for each marker.

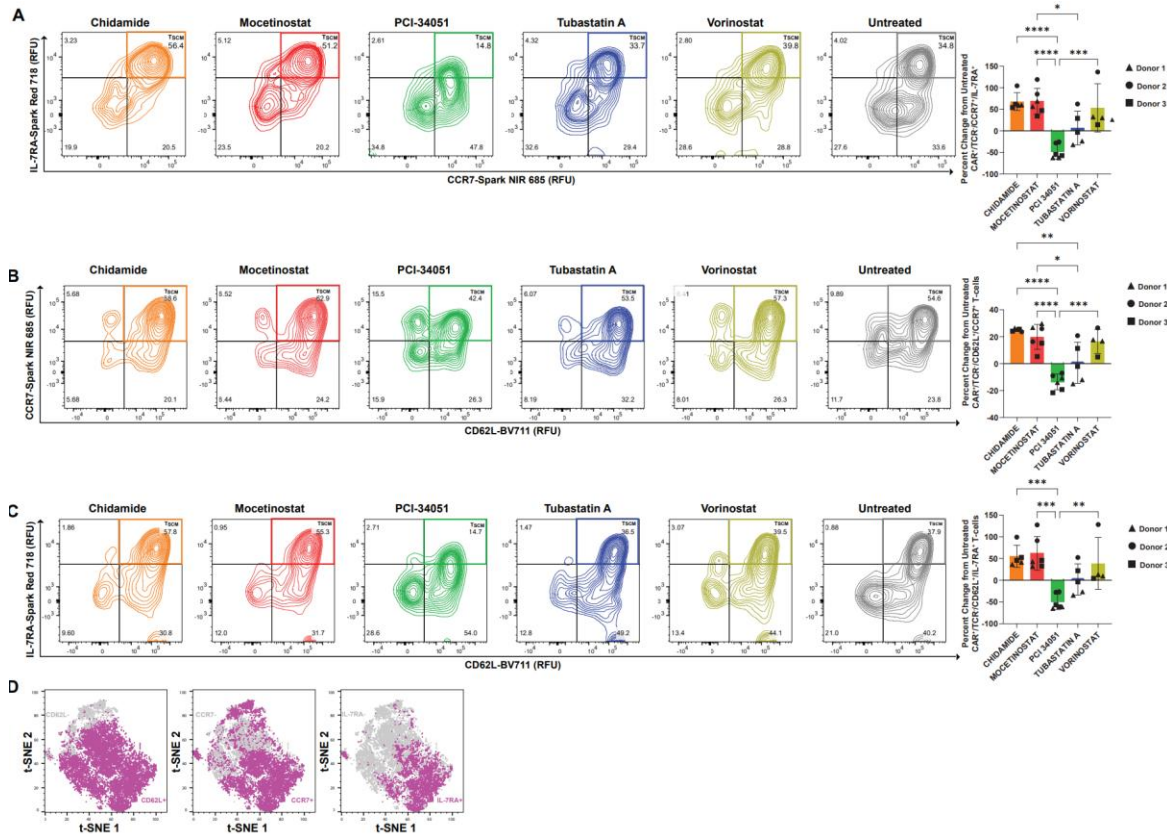


Figure S5. TscM marker expression in *TRAC*-CAR T cells treated with HDAC inhibitors. Representative contour plots and bar graphs of the percent change in double positive expression of (A) IL-7RA and CCR7, (B) CCR7 and CD62L, (C) IL-7RA and IL-7RA and in HDACi-treated *TRAC*-CAR-T cells. (D) Marker expression t-SNE plots of *TRAC*-CAR-T cells treated with Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat, or untreated cells. These plots were generated using flow cytometry expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/TCR/*CAR*⁺ T cells. The relative populations of CD62L⁺/-, CCR7⁺/-, and IL-7RA⁺/- cells are shown (Three donors, $N_{\text{Chidamide}} = 5$, $N_{\text{Mocetinostat}} = 5$, $N_{\text{PCI-34051}} = 6$, $N_{\text{Tubastatin A}} = 5$, $N_{\text{Vorinostat}} = 4$). Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. **CCR7⁺/IL-7RA⁺**: (Chidamide: 68.0% [20.7], Mocetinostat: 68.6% [30.4], PCI-34051: -48.5% [17.1], Tubastatin A: 6.4% [39.2], Vorinostat: 53.4% [56.1]) (Chidamide vs. PCI 34051: $p < 0.0001$, Mocetinostat vs. PCI 34051: $p < 0.0001$, Mocetinostat vs. Tubastatin A: $p = 0.0409$, PCI 34051 vs. Vorinostat: $p = 0.001$); **CCR7⁺/CD62L⁺**: (Chidamide: 25.1% [1.1], Mocetinostat: 19.9% [9.3], PCI-34051: -13.6% [5.8], Tubastatin A: 0.6% [15.3], Vorinostat: 16.5% [8.9]) (Chidamide vs. PCI 34051: $p < 0.0001$, Chidamide vs. Tubastatin A: $p = 0.0033$, Mocetinostat vs. PCI 34051: $p = 0.0001$, Mocetinostat vs. Tubastatin A: $p = 0.0178$, PCI 34051 vs. Vorinostat: $p = 0.0004$); **CD62L⁺/IL-7RA⁺**: (Chidamide: 55.0% [25.4], Mocetinostat: 62.0% [38.8], PCI-34051: -49.6% [17.7], Tubastatin A: 1.9% [35.8], Vorinostat: 38.7% [59.8]) (Chidamide vs. PCI 34051: $p = 0.0009$, Mocetinostat vs. PCI 34051: $p = 0.0002$, PCI 34051 vs. Vorinostat: $p = 0.0087$).

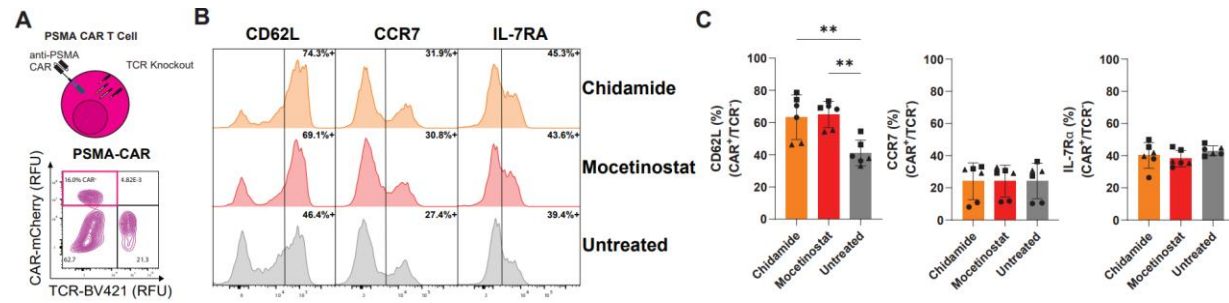


Figure S6. Phenotype of Second-Generation PSMA *TRAC*-CAR T cell with 41BB Costimulatory Domain. (A) PSMA *TRAC*-CAR T cells were manufactured over the course of 14 days. (B) Representative histograms of CD62L, CCR7, and IL-7RA in HDACi-treated GD2 *TRAC*-CAR T cells showing the relative positive percentage for each marker within the bulk population. (C) Bar graphs showing the percent change in CD62L⁺, CCR7⁺, and IL-7RA⁺ for each HDACi-treated group. Three biological donors, N_{Chidamide} = 6 data points, N_{Mocetinostat} = 6 data points, N_{Untreated} = 6 data points. Error bar represents standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; **p<0.01.

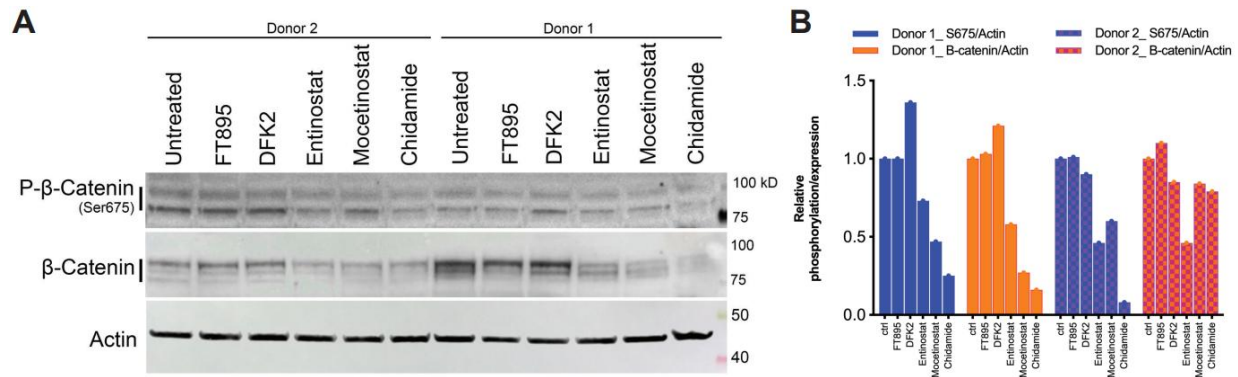


Figure S7. Total β-Catenin and P-β-Catenin (S675) Protein from HDACi Treatment. Cells were treated with HDAC inhibitors for 72 hours. Post-treatment, cells were lysed, and total protein samples were isolated. Equal amounts of protein from each condition were subjected to SDS-PAGE. **(A)** Phospho-β-Catenin (Ser675) levels were detected by Western blotting, and the blot was subsequently stripped and reprobed for total β-Catenin. Actin served as a loading control. **(B)** Histograms represent the ratio of phospho-β-Catenin or total β-Catenin to actin, normalized to DMSO-treated controls for each donor.

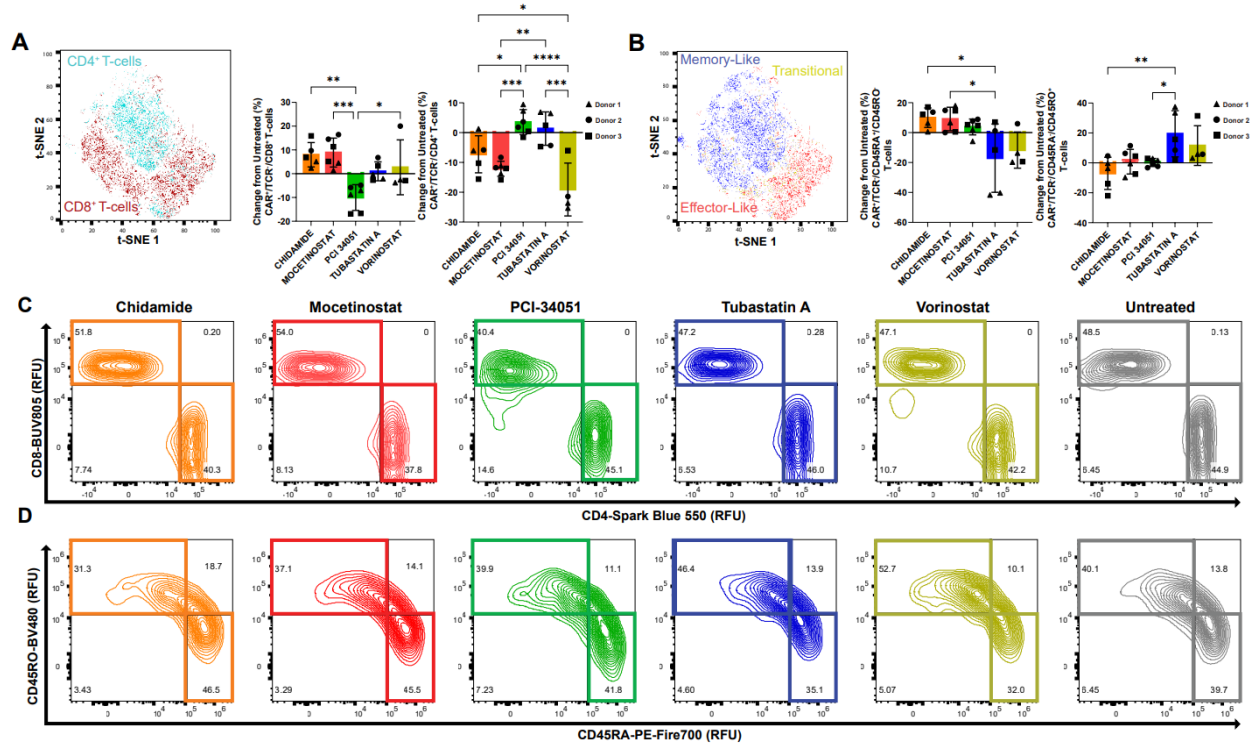


Figure S8. CD4, CD8, CD45RO, and CD45RA expression in *TRAC*-CAR T cells treated with HDAC inhibitors. Marker expression t-SNE plots of *TRAC*-CAR-T cells generated using flow cytometry expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/TCR⁺/CAR⁺ T cells showing expression of (A) CD8 and CD4 and (B) CD45RO and CD45RA and bar graphs of the percent change in positive expression relative to untreated cells for GD2 *TRAC*-CAR-T cells treated with Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat. Representative contour plots of (C) CD8 vs CD4 and (D) CD45RA vs CD45RO for HDACi treated and untreated T cells. Three donors, $N_{\text{Chidamide}} = 5$, $N_{\text{Mocetinostat}} = 5$, $N_{\text{PCI-34051}} = 6$, $N_{\text{Tubastatin A}} = 5$, $N_{\text{Vorinostat}} = 4$. Error bars represent mean and standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. **CD8⁺**: (Chidamide: 8.0% [5.1], Mocetinostat: 8.9% [6.1], PCI-34051: -10.0% [5.4], Tubastatin A: 1.0% [4.0], Vorinostat: 2.7% [11.6]) (Chidamide vs. PCI 34051: p=0.0015, Mocetinostat vs. PCI 34051: p=0.0005, PCI 34051 vs. Vorinostat: p=0.0473); **CD4⁺**: (Chidamide: -7.3% [6.2], Mocetinostat: -12.1% [2.5], PCI-34051: 3.5% [4.2], Tubastatin A: 1.4% [5.6], Vorinostat: -19.1% [8.9]) (Chidamide vs. PCI 34051: p=0.0289, Chidamide vs. Vorinostat: p=0.0313, Mocetinostat vs. PCI 34051: p=0.0006, Mocetinostat vs. Tubastatin A: p=0.0047, CI 34051 vs. Vorinostat: p<0.0001, Tubastatin A vs. Vorinostat: p=0.0002). **CD45RO⁺/CD45RA⁺**: (Chidamide: 10.0% [6.54], Mocetinostat: 9.1% [7.8], PCI-34051: 3.8% [5.3], Tubastatin A: -17.1% [22.9], Vorinostat: -11.7% [12.1]) (Chidamide vs. Tubastatin A: p=0.0166, Mocetinostat vs. Tubastatin A: p=0.0152); **CD45RO⁺/CD45RA⁻**: (Chidamide: -7.0% [10.8], Mocetinostat: 0.8% [8.2], PCI-34051: 0.3% [2.7], Tubastatin A: 19.5% [15.4], Vorinostat: 11.6% [13.3]) (Chidamide vs. Tubastatin A: p=0.0054, PCI 34051 vs. Tubastatin A: p=0.046)

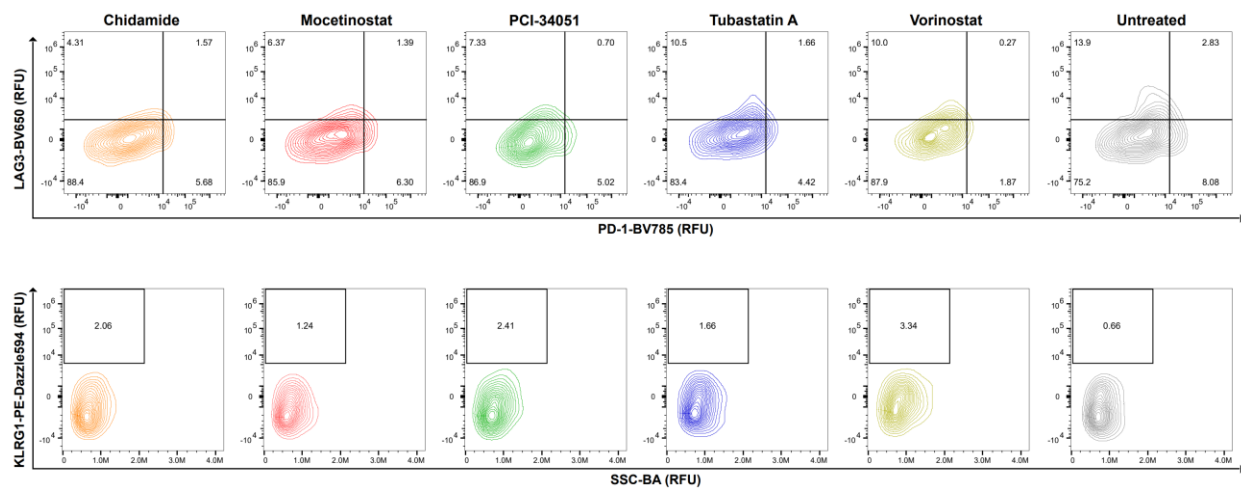


Figure S9. Exhaustion and energy surface marker expression in GD2 *TRAC*-CAR-T cells treated with HDAC inhibitors. Representative contour plots of LAG3, PD-1 and KLRG1 expression in GD2 *TRAC*-CAR T cells treated with vehicle (untreated), Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat.

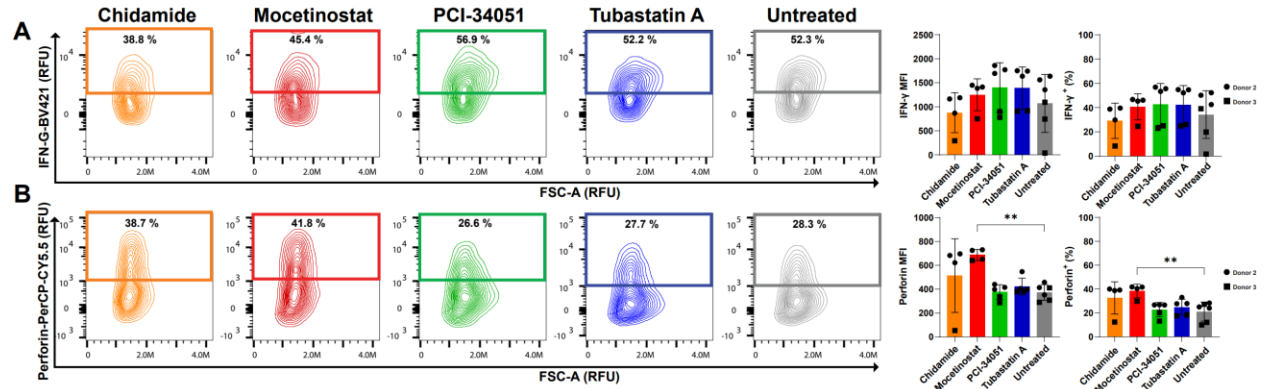


Figure S10. IFN-gamma and perforin expression in GD2 *TRAC*-CAR T cells treated with HDAC inhibitors. GD2 *TRAC*-CAR T cells were stimulated with CHLA-20 neuroblastoma cells using a 5:1 effector-to-target ratio for 24 hours, after which they were harvested for flow cytometry for internal expression of cytokines. CD45⁺/TCR⁻ T cells were analyzed for their expression of (A) IFN-gamma and (B) perforin. Representative contour plots showing the expression and bar graphs depicting the MFI and percent positivity of each marker in GD2 *TRAC*-CAR T cells treated with Chidamide, Mocetinostat, PCI-34051, Tubastatin A, or vehicle (untreated). Two donors, $N_{\text{Chidamide}} = 4$, $N_{\text{Mocetinostat}} = 4$, $N_{\text{PCI-34051}} = 5$, $N_{\text{Tubastatin A}} = 5$, $N_{\text{Untreated}} = 6$. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; ** $p < 0.01$. **IFN- γ MFI:** (Chidamide: 878 RFU [417], Mocetinostat: 1249 RFU [333], PCI-34051: 1401 RFU [517], Tubastatin A: 1391 RFU [439], Untreated: 1071 RFU [603]); **IFN- γ ⁺:** (Chidamide: 29.2% [14.5], Mocetinostat: 40.7% [10.8], PCI-34051: 42.8% [17.2], Tubastatin A: 42.6% [15.6], Untreated: 34.3% [19.7]); **Perforin MFI:** (Chidamide: 514 RFU [308], Mocetinostat: 687 RFU [47], PCI-34051: 375 RFU [62], Tubastatin A: 423 RFU [70], Untreated: 372 RFU [66]) (Untreated vs Mocetinostat: $p = 0.007$); **Perforin⁺:** (Chidamide: 32.6% [13.5], Mocetinostat: 38.4% [5.8], PCI-34051: 22.8% [6.0], Tubastatin A: 24.9% [6.6], Untreated: 21.1% [8.0]) (Untreated vs Mocetinostat: $p = 0.0097$)

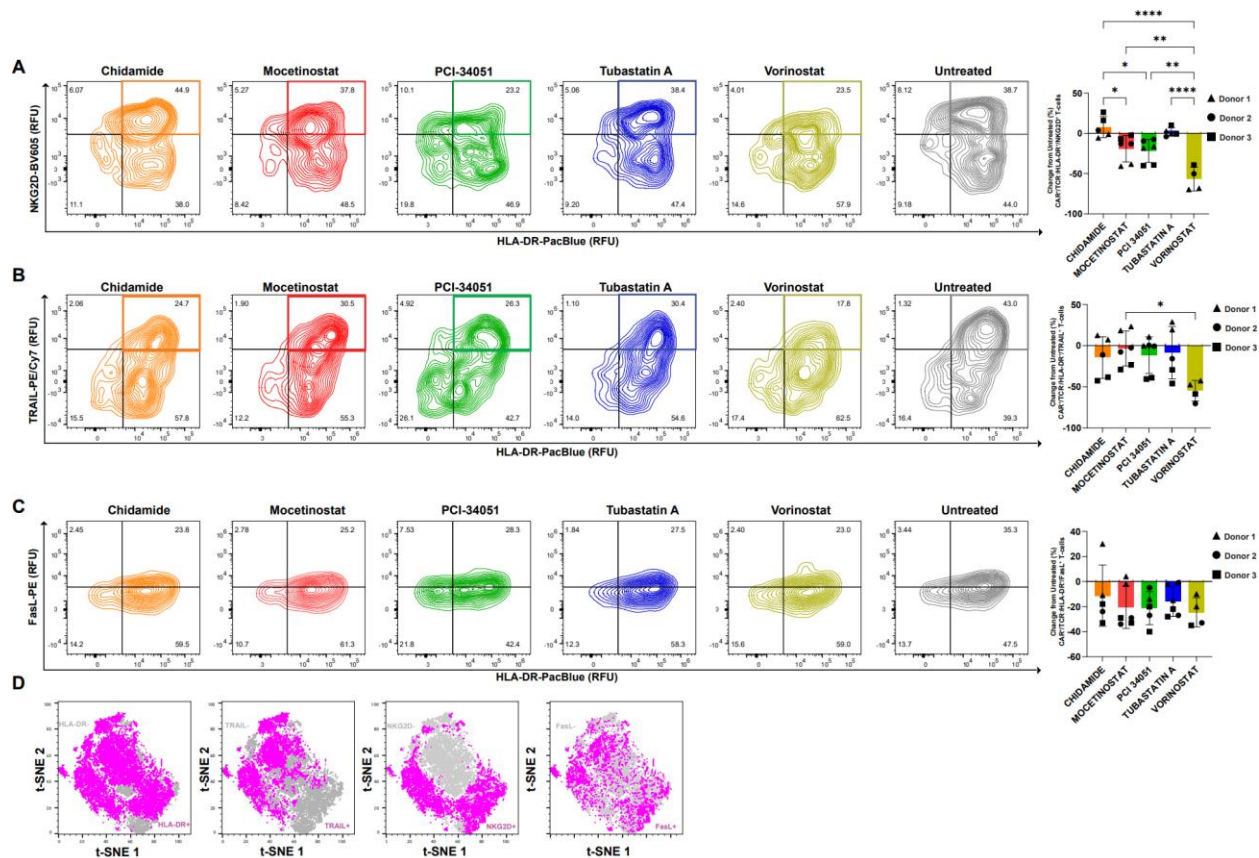


Figure S11. CAR-independent cytotoxicity marker expression in GD2 *TRAC*-CAR T cells treated with HDAC inhibitors. Representative contour plots and bar graphs of the percent change in double positive expression of (A) NKG2D and HLA-DR, (B) TRAIL and HLA-DR, (C) FasL and HLA-DR and in HDACi-treated GD2 *TRAC*-CAR T cells. (D) Marker expression t-SNE plots of *TRAC*-CAR-T cells treated with Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat, or vehicle (untreated). These plots were generated using flow cytometry expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/TCR⁺/CAR⁺ T cells. The relative populations of HLA-DR⁺/-, TRAIL⁺/-, NKG2D⁺/-, and FasL⁺/- cells are shown (Three donors, $N_{\text{Chidamide}} = 5$, $N_{\text{Mocetinostat}} = 5$, $N_{\text{PCI-34051}} = 6$, $N_{\text{Tubastatin A}} = 5$, $N_{\text{Vorinostat}} = 4$. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$. **NKG2D⁺/HLA-DR⁺**: (Chidamide: 7.8% [13.2], Mocetinostat: -19.0% [16.4], PCI-34051: -21.8% [14.4], Tubastatin A: 1.6% [5.3], Vorinostat: -56.9% [14.7]) (Chidamide vs. Mocetinostat: $p = 0.0263$, Chidamide vs. PCI 34051: $p = 0.0123$, Chidamide vs. Vorinostat: $p < 0.0001$, Mocetinostat vs. Vorinostat: $p = 0.0024$, PCI 34051 vs. Vorinostat: $p = 0.0051$, Tubastatin A vs. Vorinostat: $p < 0.0001$) **TRAIL⁺/HLA-DR⁺**: (Chidamide: -14.5% [25.3], Mocetinostat: -3.5% [21.2], PCI-34051: -11.5% [22.2], Tubastatin A: -8.6% [31.9], Vorinostat: -54.7% [12.4]) (Mocetinostat vs. Vorinostat: $p = 0.0228$) **FasL⁺/HLA-DR⁺**: (Chidamide: -11.2% [24.4], Mocetinostat: -20.5% [16.9], PCI-34051: -21.2% [13.3], Tubastatin A: -15.8% [12.0], Vorinostat: -24.5% [11.7])

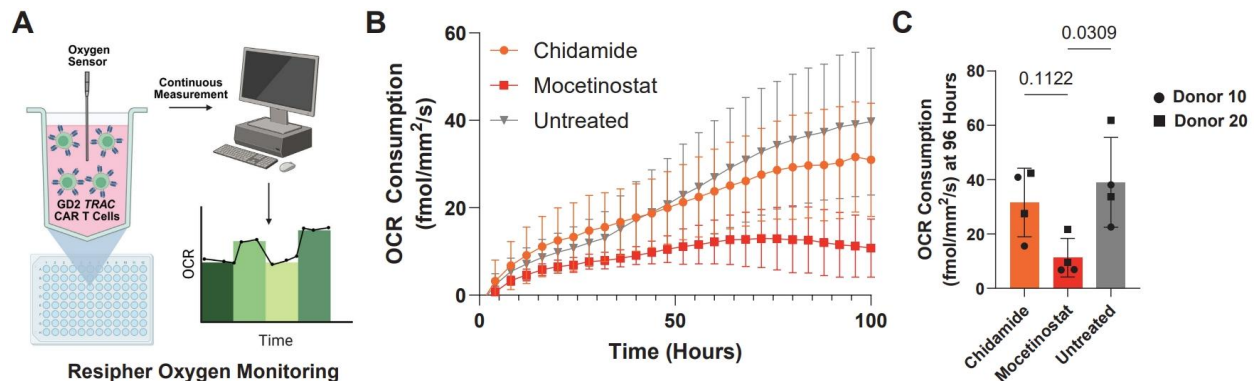


Figure S12. Mocetinostat Decreases Oxygen Consumption During Treatment. (A) We used the Resipher system to monitor continuously measure the oxygen consumption of *TRAC*-CAR T cells treated with Chidamide and Mocetinostat. (C) Dot plot of oxygen consumption over 96 hours and (D) bar of consumption at 96 hours. Two donors, $N_{\text{Chidamide}} = 4$, $N_{\text{Mocetinostat}} = 4$,. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. (Chidamide: 31.6 fmol/mm²/s [12.6], Mocetinostat: 11.3 fmol/mm²/s [7.1], Untreated: 39.1 fmol/mm²/s [16.5]) (Chidamide, Untreated vs Mocetinostat: $p = 0.11$, 0.031).

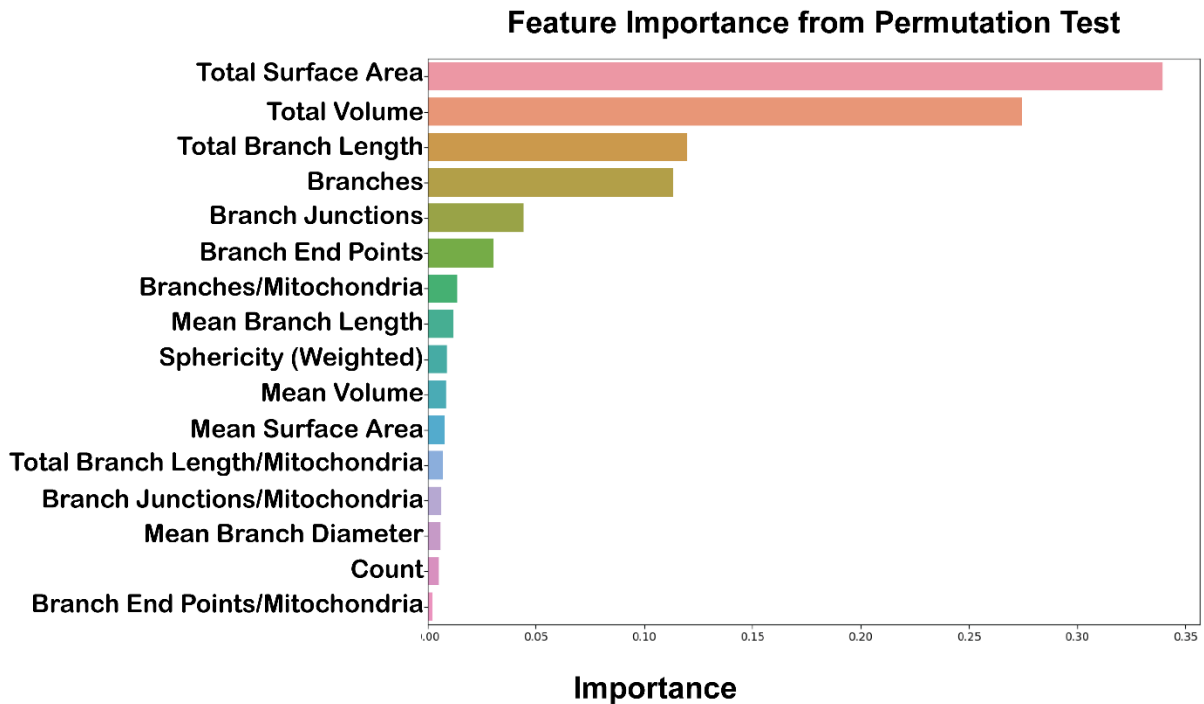
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Figure S13. Mitochondrial Morphology QC for HDACi-Treated GD2 *TRAC*-CAR T Cells.

GD2 *TRAC*-CAR T cells were treated with HDACi for 72 hours before being stained with Mitotracker Red CmxROS, imaged on a single cell level, and analyzed using Imagej to produce statistics regarding the mitochondria in each single cell image (count, total volume, mean volume, total surface area, mean surface area, sphericity, branches, branches/per mitochondria, total branch length, total branch length/per mitochondria, mean branch length, branch junctions, branch junctions/per mitochondria, branch end points, branch end points/per mitochondria, mean branch diameter). (A) Permutation test was performed to assess the importance of each feature ranked in a bar graph as shown.

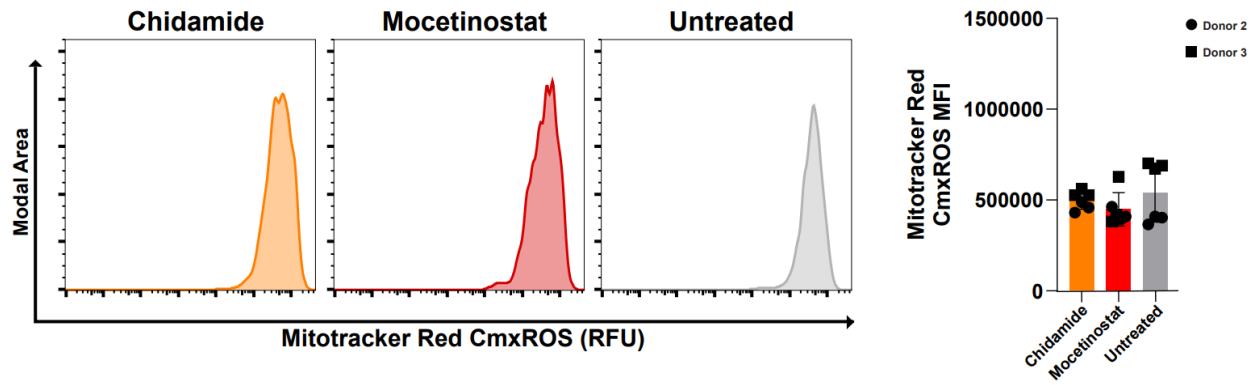
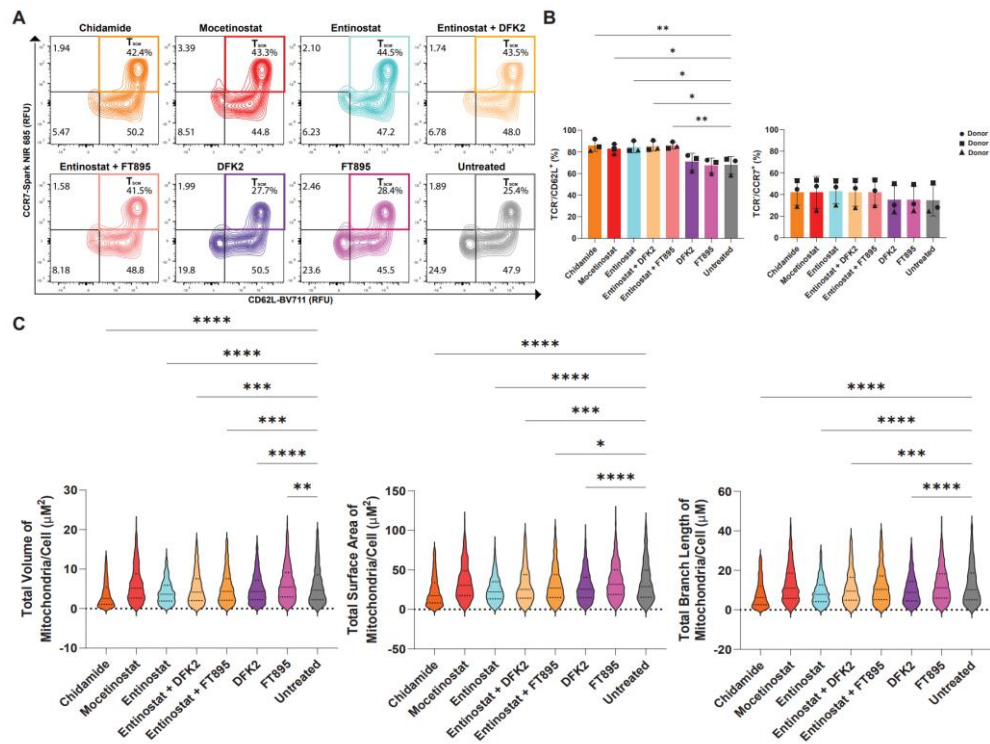


Figure S14. Mitochondrial Mass of HDACi-treated GD2 TRAC-CAR T Cells. GD2 TRAC-CAR T cells were treated with HDACi's (Chidamide, Mocetinostat, PCI-34051, Tubastatin A, Vorinostat) for 72 hours before being stained with Mitotracker Red CmxROS stain to measure mitochondrial mass. The expression on representative histograms and MFI on bar graphs are shown. Two donors, $N_{\text{Chidamide}} = 4$, $N_{\text{Mocetinostat}} = 4$, $N_{\text{PCI-34051}} = 5$, $N_{\text{Tubastatin A}} = 5$, $N_{\text{Untreated}} = 6$. Error bars represent mean and standard deviation. (Chidamide: 498833 RFU[48914], Mocetinostat: 449139 RFU [91960], Untreated: 540186 RFU [162502]).



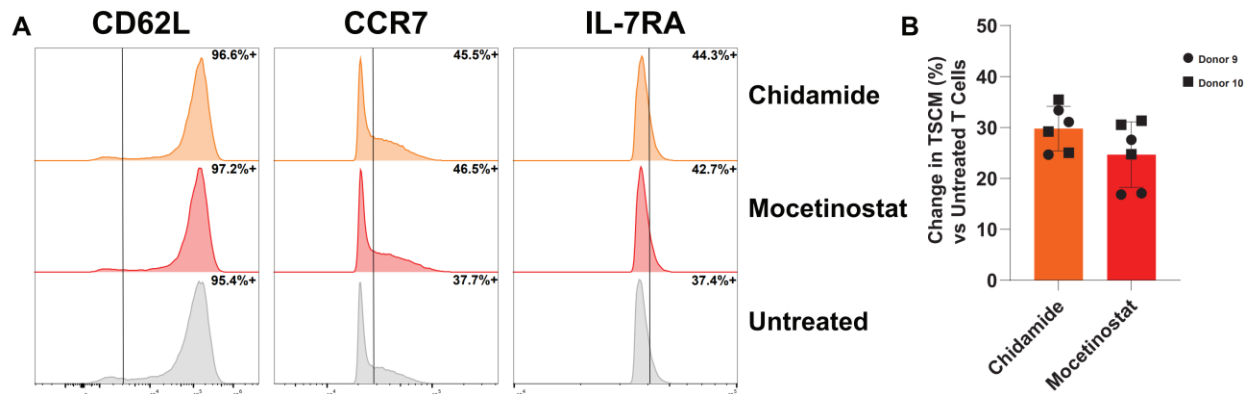


Figure S16. Chidamide and Mocetinostat Modulate T_{SCM} phenotype of Fresh GD2 *TRAC*-CAR T cells. GD2 *TRAC*-CAR T cells were manufactured over the course of 14 days and treated with HDACi for 72 hours. (A) Representative histograms of CD62L, CCR7, and IL-7RA in HDACi-treated GD2 *TRAC*-CAR T cells showing the relative positive percentage for each marker within the bulk population, (B) bar graph showing the percent change in T_{SCM} (CD62L⁺/CCR7⁺/IL-7RA⁺) relative to untreated GD2 *TRAC*-CAR T cells for each HDACi-treated group. Two biological donors, N_{Chidamide} = 6 data points, N_{Mocetinostat} = 6 Data Points. Error bar represents standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons.

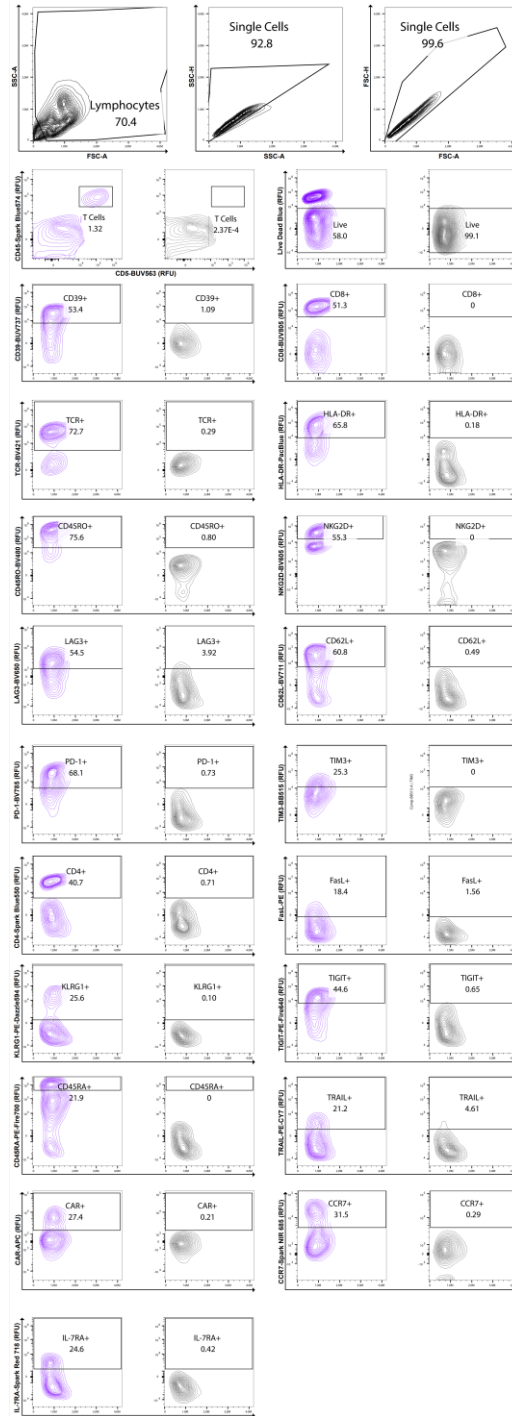


Figure S17. Gating Scheme and FMO's for *In Vivo* Spectral Flow Cytometry. Splenic lymphocytes isolated from mice were stained with an optimized antibody panel and analyzed via flow cytometry in FlowJo. Live T cell events were isolated by excluding debris, doublets, dead cells from the analysis, and filtering for CD45⁺/CD5⁺ events. Positive gates for each surface marker were identified by staining a fluorescence minus one control with each antibody except for the one of interest. The FMO was then used to isolate a positive population in a fully stained sample, both of which are displayed for each marker.

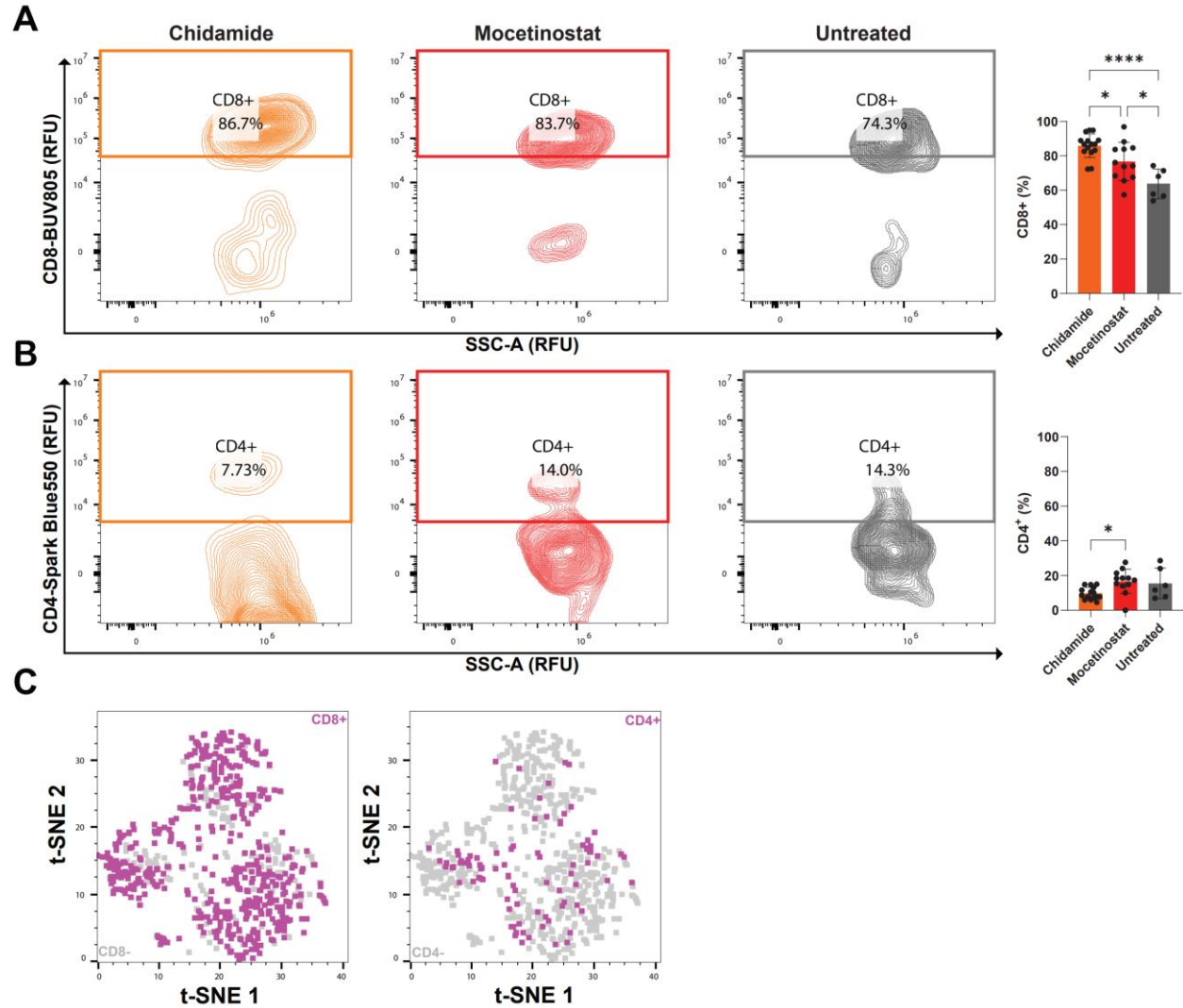


Figure S18. CD4 and CD8 Expression in Splenic Lymphocytes of HDACi-Treated GD2 *TRAC*-CAR T Cells. Representative contour plots and bar graphs of the percent change in positive expression of (A) CD8, (B) CD4 in HDACi-treated GD2 *TRAC*-CAR T cells. (C) Marker expression t-SNE plots of GD2 *TRAC*-CAR T cells treated with Chidamide, Mocetinostat or vehicle (untreated). These plots were generated using flow cytometry expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD39, TIM3, TIGIT, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/CD45⁺/CD5⁺/TCR⁻/CAR⁺ T cells. The relative +/- populations of each marker are shown One donor, $N_{\text{Chidamide}} = 15$, $N_{\text{Mocetinostat}} = 12$, $N_{\text{Untreated}} = 6$ data points. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons. * $p < 0.05$, **** $p < 0.0001$. CD8+: (Chidamide: 85.9% [6.9], Mocetinostat: 76.7% [11.1], Untreated: 63.7% [8.6]) (Chidamide vs. Mocetinostat: $p = 0.0324$, Chidamide vs. Untreated: $p < 0.0001$, Mocetinostat vs. Untreated: $p = 0.0182$); CD4+: (Chidamide: 9.7% [3.5], Mocetinostat: 16.7% [7.0], Untreated: 15.6% [8.8]) (Chidamide vs. Mocetinostat: $p = 0.0158$).

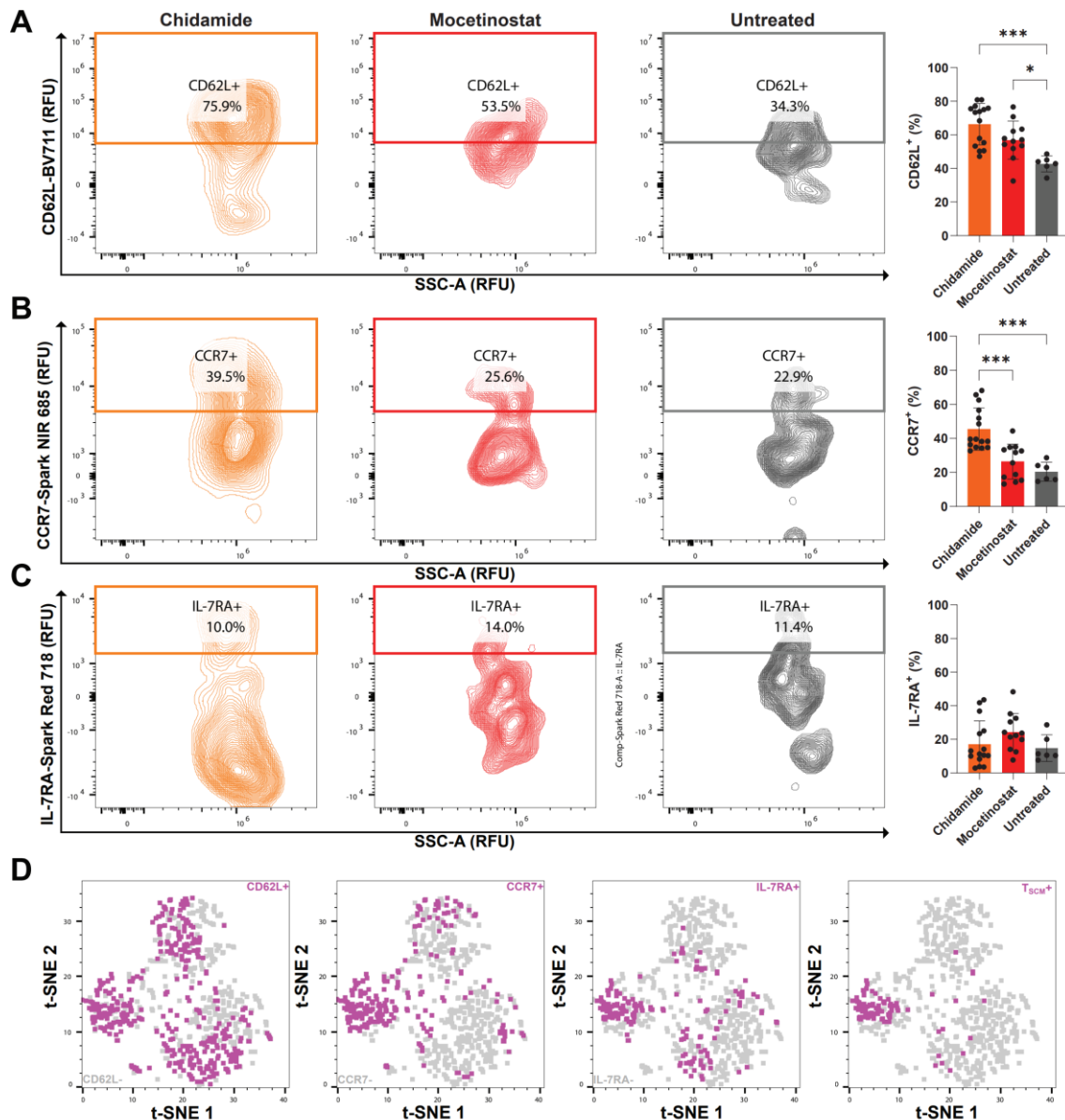
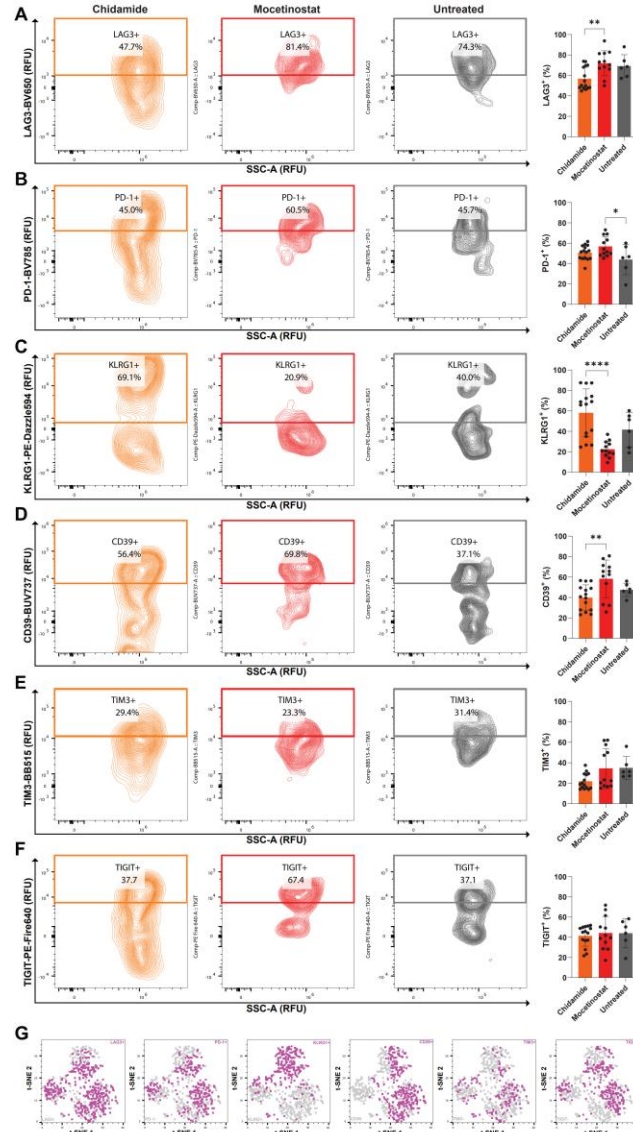


Figure S19. T_{SCM} Marker Expression in Splenic Lymphocytes of HDACi-Treated GD2 *TRAC*-CAR T Cells. Representative contour plots and bar graphs of the expression of (A) CD62L, (B) CCR7, (C) IL-7RA in Chidamide, Mocetinostat or vehicle (untreated) GD2 *TRAC*-CAR T cells. (D) Marker expression t-SNE plots of Chidamide, Mocetinostat or vehicle (untreated) GD2 *TRAC*-CAR T cells generated using flow cytometry data for expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD39, TIM3, TIGIT, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/CD45⁺/CD5⁺/TCR⁺/CAR⁺ T cells. The relative +/- populations of each marker are shown. Populations with less than 20 events were omitted from phenotypic analysis. One donor, $N_{\text{Chidamide}} = 15$, $N_{\text{Mocetinostat}} = 12$, $N_{\text{Untreated}} = 6$ data points. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; * $p < 0.05$, *** $p < 0.001$. CD62L: (Chidamide: 66.4% [12.4], Mocetinostat: 56.9% [11.3], Untreated: 42.7% [4.8]) (Chidamide vs. Untreated: $p = 0.0003$, Mocetinostat vs. Untreated: $p = 0.0398$); CCR7: (Chidamide: 45.4% [12.4], Mocetinostat: 26.4% [10.3], Untreated: 20.4% [5.6]); (Chidamide vs. Mocetinostat: $p = 0.0002$, Chidamide vs. Untreated: $p = 0.0001$); IL-7R α : (Chidamide: 17.2% [13.7], Mocetinostat: 24.3% [11.1], Untreated: 14.8% [8.0]).



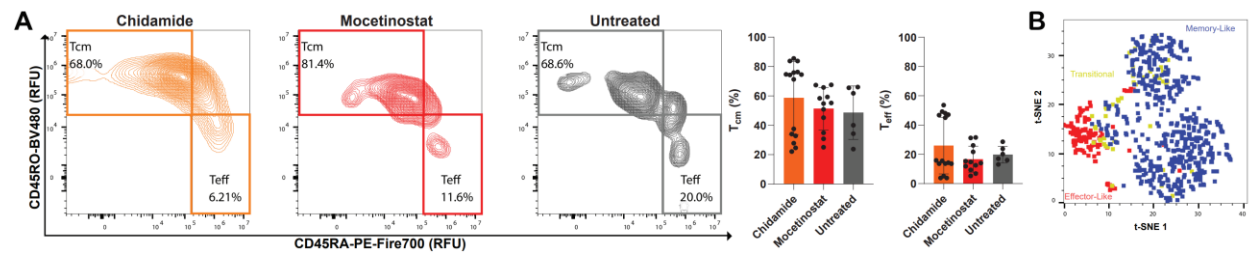


Figure S21. CD45RO/CD45RA Expression in Splenic Lymphocytes of HDACi-Treated GD2 *TRAC*-CAR T Cells. Representative contour plots and bar graphs of the percent change in positive expression of (A) CD45RO and CD45RA in HDACi-treated GD2 *TRAC*-CAR T cells. (B) Marker expression t-SNE plots of Chidamide, Mocetinostat or vehicle (untreated) GD2 *TRAC*-CAR T cells generated using flow cytometry data for expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD39, TIM3, TIGIT, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/CD45⁺/CD5⁺/TCR⁺/CAR⁺. The relative +/- populations of each marker are shown. Populations with less than 20 events were omitted from phenotypic analysis. One donor, $N_{\text{Chidamide}} = 15$, $N_{\text{Mocetinostat}} = 12$, $N_{\text{Untreated}} = 6$ data points. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons. Memory-like: (Chidamide: 58.6% [24.9], Mocetinostat: 51.3% [14.4], Untreated: 48.6% [18.2]); Effector-like: (Chidamide: 26.0% [19.4], Mocetinostat : 16.7% [8.9], Untreated: 19.8% [5.7]).

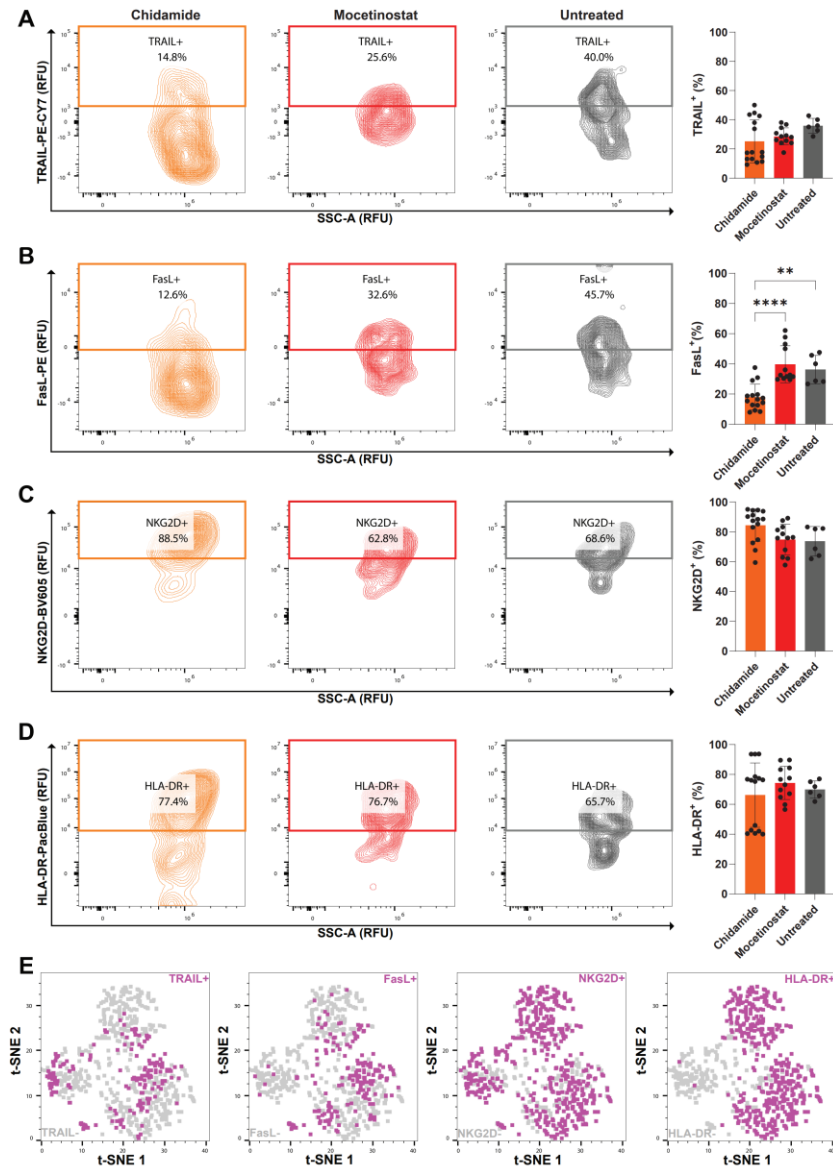


Figure S22. TRAIL, FasL, NKG2D, and HLA-DR Expression in Splenic Lymphocytes of HDACi-Treated GD2 *TRAC*-CAR T Cells. Representative contour plots and bar graphs of the percent change in positive expression of (A) TRAIL, (B) FasL, (C) NKG2D, (D) HLA-DR, in HDACi-treated GD2 *TRAC*-CAR T cells. (E) Marker expression t-SNE plots of Chidamide, Mocetinostat or vehicle (untreated) GD2 *TRAC*-CAR T cells generated using flow cytometry data for expression of CD45RO, CD45RA, LAG3, PD-1, KLRG1, CD39, TIM3, TIGIT, CD62L, CCR7, IL-7RA, NKG2D, TRAIL, FASL, and HLA-DR on Live/CD45⁺/CD5⁺/TCR⁻/CAR⁺ T cells. The relative +/- populations of each marker are shown. One donor, $N_{\text{Chidamide}} = 15$, $N_{\text{Mocetinostat}} = 12$, $N_{\text{Untreated}} = 6$ data points. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; ** $p < 0.01$, **** $p < 0.0001$. TRAIL: (Chidamide: 25.2% [14.9], Mocetinostat: 28.6% [5.8], Untreated: 35.9% [5.1]); NKG2D: (Chidamide: 84.4% [10.9], Mocetinostat: 74.8% [10.2], Untreated: 73.6% [9.9]); HLA-DR: (Chidamide: 66.2% [21.6], Mocetinostat: 74.3% [11.1], Untreated: 69.9% [5.8]); FasL: Chidamide: 18.0% [8.5], Mocetinostat: 39.7% [12.2], Untreated: 36.2% [9.4]) (Chidamide vs. Mocetinostat: $p < 0.0001$, Chidamide vs. Untreated: $p = 0.0024$).

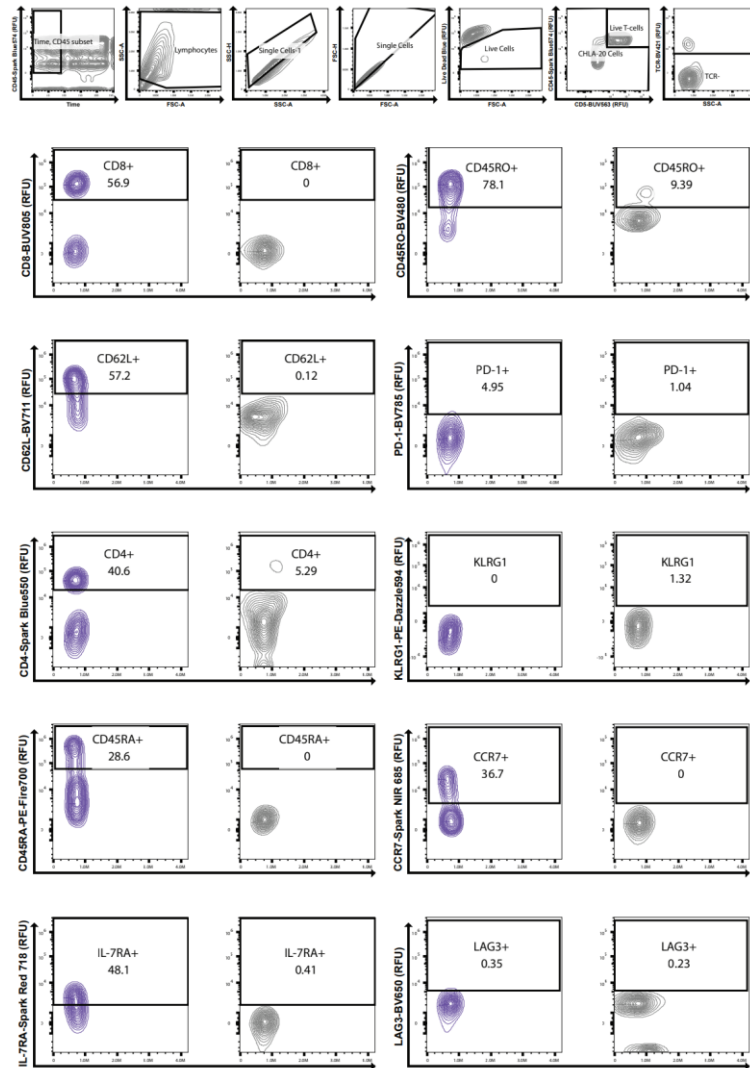


Figure S23. Gating Scheme and FMO's for Serial Stimulation Flow Cytometry. GD2 *TRAC*-CAR T cell and CHLA-20 cocultures were stained with an optimized antibody panel and analyzed via flow cytometry in FlowJo.

Live T cell events were isolated by excluding debris, doublets, dead cells from the analysis, and filtering for CD45⁺/CD5⁺ events. Positive gates for each surface marker were identified by staining a fluorescence minus one control with each antibody except for the one of interest. The FMO was then used to isolate a positive population in a fully stained sample, both of which are displayed for each marker.

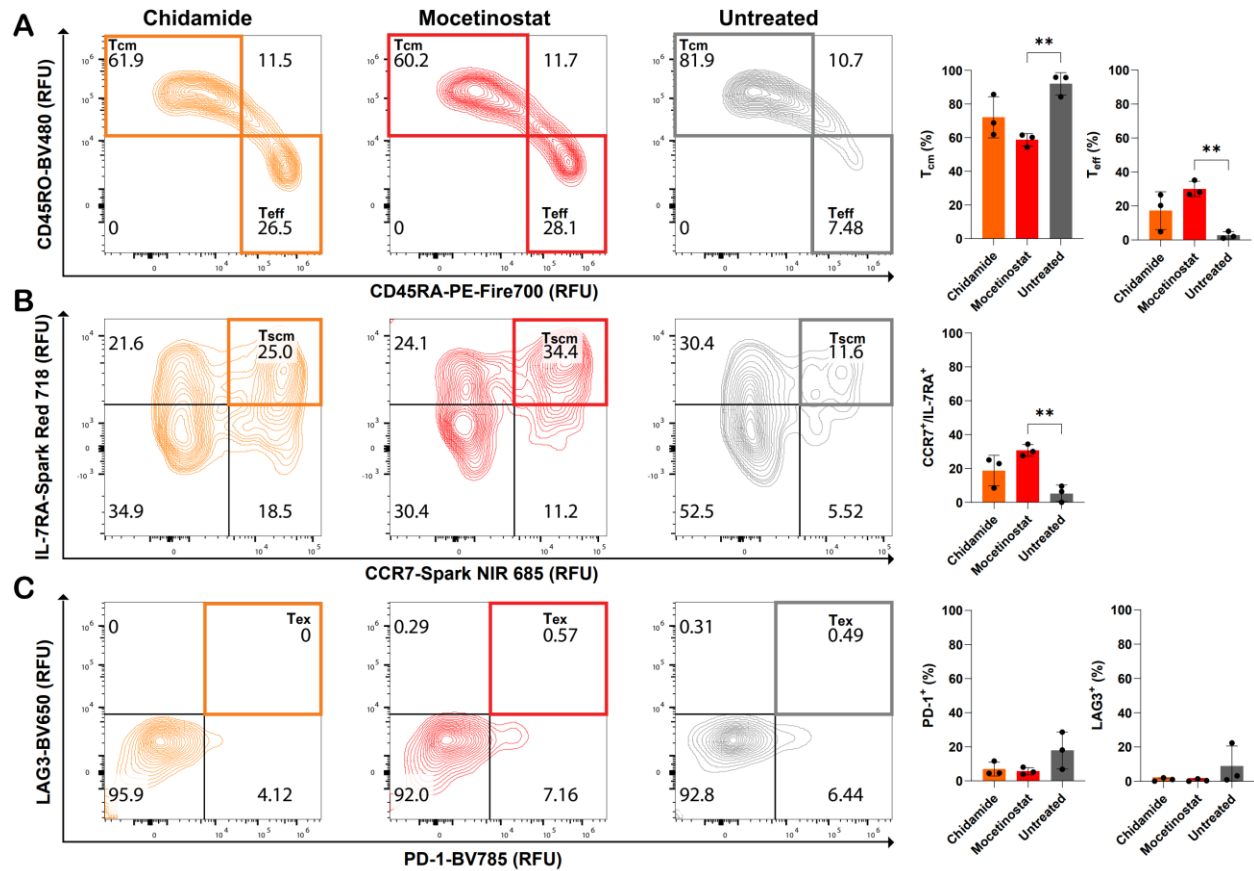


Figure S24. Phenotypic Analysis of Serially Stimulated HDACi-treated GD2 TRAC-CAR T cells. GD2 TRAC-CAR T cells were pre-treated with Chidamide, Mocetinostat, or no treatment for 72 hours and then serially stimulated with CHLA-20 neuroblastoma cells at a 2:1 effector-to-target ratio for 10 days with new cells added every 2-3 days. Samples were then stained with an optimized antibody panel and analyzed via spectral flow cytometry. Representative contour plots and bar graphs show the expression of cell populations with (A) CD45RO and CD45RA, (B) IL-7RA and CCR7, and (C) LAG3 and PD-1. One donor, $N_{\text{Chidamide}} = N_{\text{Mocetinostat}} = N_{\text{Untreated}} = 3$ data points. Error bars represent standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; $**p < 0.01$. Effector-like: (Chidamide: 17.2% [11.1], Mocetinostat: 30.1% [4.5], Untreated: 2.7% [2.1]) (Mocetinostat vs. Untreated: $p = 0.0075$); Memory-like: (Chidamide: 72.1% [12.3], Mocetinostat: 58.7% [3.8], Untreated: 92.0% [6.6]) (Mocetinostat vs. Untreated: $p = 0.0066$); T_{SCM}: (Chidamide: 18.8% [9.0], Mocetinostat: 30.7% [3.4], Untreated: 5.3% [4.9]) (Mocetinostat vs. Untreated: $p = 0.006$); PD-1 (Chidamide: 6.9% [4.1], Mocetinostat: 5.7% [2.1], Untreated: 17.8% [10.9]); LAG3 (Chidamide: 1.0% [0.9], Mocetinostat: 0.6% [0.7], Untreated: 8.8% [11.8]).

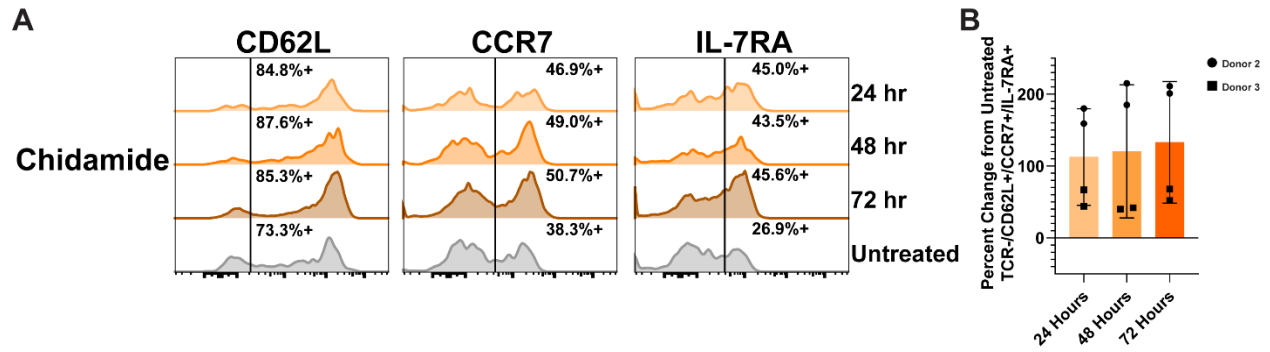


Figure S25. HDAC Inhibitors Modulate T_{SCM} phenotype of GD2 TRAC-CAR T cells at Different Time Points.

GD2 TRAC-CAR T cells were manufactured over the course of 14 days and cryopreserved. After thaw, they were rested for 24 hours before treatment with HDACis for 72 hours. (A) Representative histograms of CD62L, CCR7, and IL-7RA in HDACi-treated GD2 TRAC-CAR T cells showing the relative positive percentage for each marker within the bulk population and (B) bar graph showing the percent change in T_{SCM} (CD62L⁺/CCR7⁺/IL-7RA⁺) relative to untreated GD2 TRAC-CAR T cells for each HDACi-treated group. Two biological donors, $N_{\text{Chidamide}} = 4$ data points. Error bar represents standard deviation. Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons.

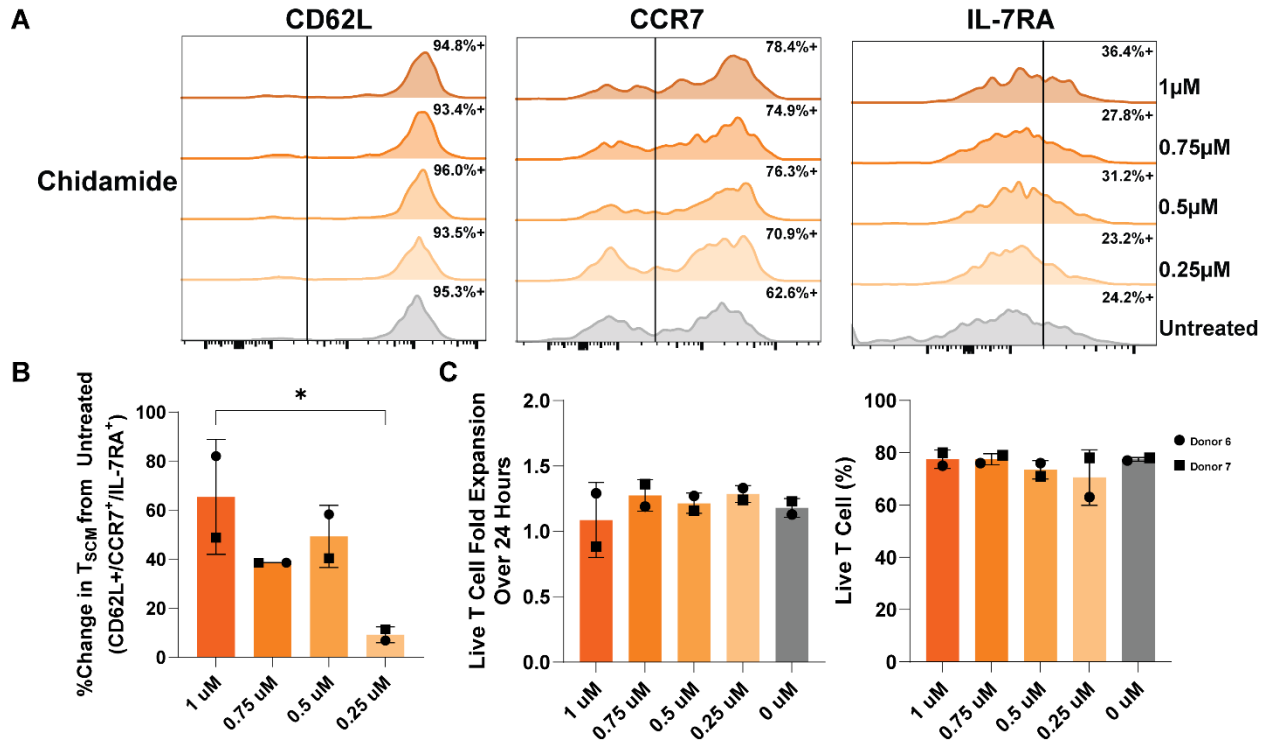


Figure S26. Chidamide Modulates T_{SCM} phenotype of Fresh GD2 TRAC-CAR T cells after 24 Hours at Different Concentrations. GD2 TRAC-CAR T cells were manufactured over the course of 14 days and treated with HDACis for 24 hours. (A) Representative histograms of CD62L, CCR7, and IL-7RA in HDACi-treated GD2 TRAC-CAR T cells showing the relative positive percentage for each marker within the bulk population, (B) bar graph showing the percent change in T_{SCM} (CD62L⁺/CCR7⁺/IL-7RA⁺) relative to untreated GD2 TRAC-CAR T cells for each HDACi-treated group, and (C) bar graphs showing the percent of live T cells and fold-expansion during HDACi treatment. Two biological donors, N_{Chidamide} = 2 data points. Error bar represents standard deviation.

Statistical significance was determined with One Way ANOVA tests using Dunnett's T3 test for multiple comparisons; *p<0.05. (Chidamide): (1 μ M: 65 [23]%, 0.75 μ M: 39.0 [0.1]%, 0.5 μ M: 49.3 [13]%, 0.25 μ M: 9 [3]%) (1 μ M vs 0.25 μ M: p = 0.046).

Table S1. HDAC Inhibitors Information for Targeted HDAC Class and Isoform and Concentration Used.

Each HDAC inhibitor: Chidamide, Mocetinostat, PCI-34051, Tubastatin A, and Vorinostat target specific HDAC isoforms which separated into four class (I, IIa, IIb, and IV), which are listed below. The concentration chosen from the IC₅₀ curves used for the duration of the study is also shown.

[illegible]

Table S2. Antibodies Used in this Study for Flow Cytometry. A table of each antibody, clone, amount used for *in vitro* staining, serial stimulation, or *in vivo* splenic lymphocyte analysis and vendor is shown below for the optimized spectral flow cytometry panel.

Antibody	Clone	Fluorophore	Volume Used	Cat No	Company
GhostRed™780	-	-	-	13-0865-T100	Tonbo Biosciences
Live/Dead Blue	-	-	-	L34961A	ThermoFisher
CD45RA	HI100	PE/Fire 700	2.5	L23105	BioLegend
CD45RO	UCHL1	BV480	1.25	304172	BD Biosciences
CD62L	DREG-56	BV711	2.5	566143	BioLegend
CCR7	G043H7	Spark NIR 685	2.5	304860	BioLegend
IL-7RA	A019D5	Spark Red 718	2.5	353258	BioLegend
LAG3	11C3C65	BV650	4	369316	BioLegend
PD-1	EH12.2H7	BV785	2.5	329929	BioLegend
KLRG1	14C2A07	PE/Dazzle 594	2.5	368608	BioLegend
Alpha/Beta TCR	IP26	APC/Fire 750	2.5	306736	BioLegend
CAR	Anti-1A7	APC	0.2	NCI, 705-0010	NCI, Novus Biologics
CD45	2D1	Spark Blue 574	0.625	368558	BioLegend
CD5	UCHT2	BUV563	0.313	741354	BD Biosciences
CD8	SK1	BUV805	0.313	612890	BD Biosciences
CD4	SK3	SB550	0.625	344656	BioLegend
NKG2D/CD314	1D11	BV605	2.5	320832	BioLegend
FasL	NOK-1	PE	2.5	306407	BioLegend
TRAIL	RIK-2	PE/Cy7	1.25	308216	BioLegend
HLA-DR	L243	PacBlue	1.25	307633	BioLegend
TIM3	7D3	BB515	1.25	565568	BD Biosciences
TIGIT	A15153G	PE-Fire640	2.5	372744	BioLegend
CD39	TU66	BUV737	0.63	612852	BD Biosciences
IFN-γ	4.SB3	BV421	2.5	502532	BioLegend
Perforin	dG9	PerCP-Cy5.5	2.5	308114	Biolegend